

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	Yuvasri.B	Reg. No.:	192421231
Section / Batch:	BTech IT	Date:	22/08/26

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Industry Problem Solving Task

40. Student: YUVASRI B

Register Number: 192421231

Q40. Smart Home Energy Management Optimization (Greedy + DP)

A smart home system must optimize energy usage across appliances to reduce electricity costs. Analyze how

optimization algorithms can be applied. Identify constraints such as user preferences and energy pricing.

Design a solution using greedy and dynamic programming techniques. Discuss scalability for multiple devices. Evaluate performance efficiency. Interpret results in energy-saving systems.

Q40. Smart Home Energy Management Optimization (Greedy + DP)

Introduction

Smart home energy management aims to reduce electricity consumption and cost while maintaining user comfort. Optimization algorithms can decide when and how appliances should operate based on energy demand, electricity prices, appliance priorities, and user preferences. Greedy algorithms provide fast decisions, while Dynamic Programming (DP) can find an optimal schedule under defined constraints.

Problem Analysis

The system manages appliances such as air conditioners, washing machines, refrigerators, water heaters, and lights. The main objective is to minimize electricity cost and unnecessary energy consumption.

Constraints

- **Energy limit:** Total energy consumption should remain within the available limit.
- **Electricity pricing:** Appliances should preferably operate during low-tariff periods.
- **User preferences:** Important appliances may need to operate at specific times.
- **Appliance priority:** Critical devices should receive higher priority.
- **Operating duration:** Some appliances require a fixed number of operating hours.
- **Comfort requirements:** Temperature and other user comfort conditions must be maintained.

Greedy Approach

The greedy method selects the best immediate option at each step. Appliances can be ranked according to energy consumption, priority, and current electricity price.

For example:

1. Calculate the cost of running each appliance.
2. Identify low-price time slots.
3. Schedule high-priority appliances first.
4. Move flexible appliances to cheaper time slots.
5. Continue until the energy constraint is satisfied.

The greedy approach is simple and fast, making it suitable for real-time energy management. However, it may not always produce the globally optimal schedule.

Dynamic Programming Approach

Dynamic Programming considers different combinations of appliances and time slots to obtain an optimal schedule.

Let:

$DP[i][e]$ = minimum electricity cost for scheduling the first i appliances using e units of energy.

The recurrence can be represented as:

$$DP[i][e] = \min(DP[i-1][e], DP[i-1][e-w_i] + cost_i)$$

where:

- i = appliance number
- e = available energy
- w_i = energy required by appliance i
- $cost_i$ = electricity cost of operating appliance i

The final DP value gives the minimum possible energy cost while satisfying the constraints.

Greedy vs Dynamic Programming

Feature	Greedy	Dynamic Programming
Decision	Local best choice	Considers multiple choices
Speed	Very fast	Higher computation
Optimality	Not always optimal	Optimal for a well-defined model
Memory	Low	Higher
Scalability	Excellent	Moderate
Suitable for	Real-time control	Detailed optimal scheduling

Scalability for Multiple Devices

For a small number of appliances, DP can efficiently determine an optimal schedule. As the number of devices and time slots increases, the DP state space becomes larger and requires more memory and computation.

For large smart homes, a **hybrid approach** can be used. Greedy scheduling can quickly handle routine decisions, while DP can optimize important or flexible appliances. Devices can also be grouped into zones or categories to reduce computational complexity.

Performance Evaluation

The performance of the system can be evaluated using:

- Total electricity cost
- Total energy consumption
- Percentage of energy saved
- Peak-hour energy reduction
- Execution time
- Number of appliances successfully scheduled
- User comfort satisfaction

Greedy algorithms provide faster execution, while DP generally provides better cost optimization when the problem size is manageable.

Interpretation of Results

The optimized system shifts flexible appliance usage from expensive peak periods to cheaper periods. This reduces electricity bills and peak energy demand while satisfying important user requirements. DP provides better global optimization, whereas the greedy method provides faster real-time decisions.

Conclusion

Smart home energy management can effectively use Greedy and Dynamic Programming techniques to reduce electricity costs and energy consumption. Greedy algorithms are efficient and scalable for quick decisions, while DP provides more optimal scheduling by considering different combinations of appliances and time slots. A hybrid Greedy-DP approach can provide a practical solution for large-scale smart homes, improving energy efficiency, reducing peak demand, and maintaining user comfort.

Code of Conduct

I Yuvasr.B / 192421231 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Yuvasri .B

RUBRICS FOR CALCULATION**Industry Problem Solving Task**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				

Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____